Topic 1 - Real sequences

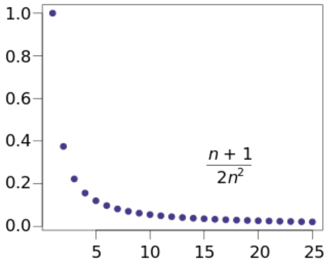
Calculus

• **Definition (Real sequence).** A real valued function is called a real sequence if its domain is the set of natural numbers.

• Shortly, functions of the form $f : \mathbb{N} \to \mathbb{R}$ are real sequences.

• For a real sequence we shall use $a, b, c, \ldots$ and, for the function $a : \mathbb{N} \to \mathbb{R}$ we use the notation $(a_n) \subseteq \mathbb{R}$.

• For a given index $n \in \mathbb{N}$, let $a_n := a(n)$. This real number is called the nth member of the sequence (a_n).



- **Examples.**
  - ⋆ *Arithmetic progression* and *geometric progression*.

    If $\lambda, d \in \mathbb{R}$ is fixed, then let $a_n := \lambda + (n-1)d$ whenever $n \in \mathbb{N}$.

    If $\lambda, q \in \mathbb{R}, q \neq 0$ is fixed, then let $a_n := \lambda q^{n-1}$ whenever $n \in \mathbb{N}$.
  - ⋆ *Harmonic sequence*. The sequence of the reciprocals of the natural numbers forms a harmonic sequence.

$$a_n := \frac{1}{n} \text{ if } n \in \mathbb{N}.$$

  - ⋆ Fibonacci's sequence: $a_1 = 1$, $a_2 = 1$, $a_3 = 2$, $a_4 = 3$, $a_5 = 5$, ...

Boundedness

• **Definition (Boundedness of real sequences).** We say that a real sequence (a_n) is *bounded from below* if there exists $k \in \mathbb{R}$ such that $a_n \geq k$ holds for all $n \in \mathbb{N}$. Similarly, we say that (a_n) is *bounded from above* if there exists $K \in \mathbb{R}$ for which $a_n \leq K$ for all $n \in \mathbb{N}$.

If (a_n) is bounded from below and from above simultaneously, then it will be called *bounded*.

• **Geometrical meaning.** Note that if (a_n) is bounded from below and from above, then there exists a bounded interval which contains the set

$$\{a_n \mid n \in \mathbb{N}\}$$

• **Exercise.** Show that the sequences defined by $a_n := \frac{1}{n^2}$ and $b_n := (-1)^n$ are bounded. Is the sequence $c_n := (-2)^n$ bounded or not?

• **Definition (Monotonicity of real sequences).** We say that (a_n) is *decreasing* or *strictly decreasing* if, for all $n \in \mathbb{N}$, we have

$$a_n \geq a_{n+1} \quad \text{or} \quad a_n > a_{n+1},$$

respectively. Similarly, (a_n) is *increasing* or *strictly increasing* if, for all $n \in \mathbb{N}$, we have

$$a_n \leq a_{n+1} \quad \text{or} \quad a_n < a_{n+1},$$

respectively.

• A sequence (a_n) is increasing and decreasing if and only if it is *constant*, that is, there exists $c \in \mathbb{R}$ such that $a_n = c$ holds for all $n \in \mathbb{N}$. Give an example of a real sequence, which is neither increasing nor decreasing!

• **Definition (Convexity of real sequences).** We say that (a_n) is *convex* if, for all $n \in \mathbb{N}$ with $n \geq 2$, we have

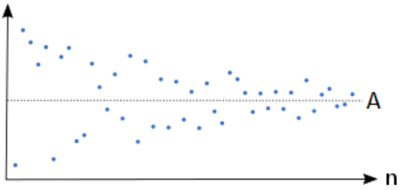
$$a_n \leq \frac{a_{n-1} + a_{n+1}}{2}.$$

The sequence (a_n) is called *concave* if the sequence $(-a_n)$ is convex.

• Show that a real sequence (a_n) is convex and concave (i.e, is affine) if and only if it is an arithmetic progression.

Convergence

• **Definition (Convergence of a real sequence).** We say that (a_n) is *convergent* if there exists $A \in \mathbb{R}$ such that, for all $\varepsilon > 0$ on can find $n \in \mathbb{N}$ such that $|a_k - A| < \varepsilon$ holds whenever $k \in \mathbb{N}$ with $k \geq n$.
Then the real number A is called a *limit* of the sequence (a_n).



• **Theorem (Uniqueness of the limit)*.** If (a_n) is convergent then it has at most one limit.
• Motivated by the above theorem, we can introduce the following notation:

$$\lim_{n\to\infty} a_n.$$

If (a_n) is convergent and its limit is A, then we shall say that (a_n) *converges* to A *as n tends to infinity* and use the notation

$$\lim_{n\to\infty} a_n = A \qquad \text{or} \qquad a_n \to A \text{ as } n \to \infty.$$

• **Definition (Divergence of a real sequence).** We say that (a_n) is *divergent* if it is not convergent. Furthermore, we shall say that

1. (a_n) *diverges to* $-\infty$ if, for all $\lambda \in \mathbb{R}$ one can find $n \in \mathbb{N}$ such that $a_k \leq \lambda$ whenever $k \geq n$, or
2. (a_n) *diverges to* $+\infty$ if, for all $\lambda \in \mathbb{R}$ one can find $n \in \mathbb{N}$ such that $a_k \geq \lambda$ whenever $k \geq n$.

If (a_n) enjoys exactly one of the properties 1. or 2. we say that (a_n) is *convergent in the extended sense*.

• **Theorem (Convergence and boundedness).*** If (a_n) is convergent, then it is bounded.

• The converse of the above theorem is not true! More precisely, there exist divergent real sequences, which are bounded.

Question: What to assume which together with boundedness implies convergence?

• **Theorem (Convergence of bounded monotone sequences).*** If (a_n) is decreasing or increasing and bounded from below or from above, respectively, then it is convergent and

$$\lim_{n\to\infty} a_n = \inf\{a_n \mid n \in \mathbb{N}\} \qquad \text{or} \qquad \lim_{n\to\infty} a_n = \sup\{a_n \mid n \in \mathbb{N}\},$$

respectively.

• **Definition (Cauchy property).** We say that (a_n) enjoys the Cauchy property or, shortly, (a_n) is a *Cauchy-sequence* if, for all $\varepsilon > 0$ we have $n \in \mathbb{N}$ such that $|a_k - a_m| < \varepsilon$ whenever $k, m \geq n$.

• **Theorem (Characterization of convergence).** A real sequence (a_n) is convergent if and only if it is a Cauchy sequence.

- ⋆ In the case of real sequences, convergence and Cauchy property are equivalent notions.
- ⋆ Advantage of the above definition: We can apply it to test the convergence of the given sequence even if we do not have any information about the exact value of the limit.

• **Definition (Subsequence).** The sequence (b_n) is called a subsequence of (a_n) if one can find (φ_n) being strictly increasing such that

$$b_n = a_{\varphi_n}, \qquad n \in \mathbb{N}.$$

• **Theorem (Convergence of a real sequence).** If (a_n) is convergent then, for all strictly increasing sequence (φ_n), the sequence (a_{φ_n}) is convergent and

$$\lim_{n\to\infty} a_{\varphi_n} = \lim_{n\to\infty} a_n.$$

- ⋆ A real sequence is not necessarily convergent even if it has convergent subsequences!
- ⋆ Each sequence is a subsequence of itself. (Indeed, let (φ_n) be the identity, that is, $\varphi_n = n$ for all $n \in \mathbb{N}$.)
- ⋆ If (a_n) is a sequence such that (a_{2n}) and(a_{2n-1}) are convergent tending to the same limit, then (a_n) is convergent and tends to the common limit.
- ⋆ If a sequence is bounded, then all of its subsequences are bounded. Consequently, if a sequence has an unbounded subsequence, then the sequence in question must be unbounded.

Accumulation points, density points

- **Definition (Accumulation points, density points).** We say that an extended real number $p \in \mathbb{R} \cup \{-\infty, +\infty\}$ is an accumulation point or a density point of (a_n) if either
  - ⋆ $p \in \mathbb{R}$ and there exists a subsequence of (a_n) which converges to p or
  - ⋆ $p \in \{-\infty, +\infty\}$ and there exists a subsequence of (a_n), which diverges to p as n tends to infinity.

For the set of accumulation points of a sequence (a_n) we shall use the notation $\mathrm{acc}(a_n)$.

- Note that if (a_n) is convergent, then

$$\lim_{n\to\infty} a_n \in \mathrm{acc}(a_n),$$

furthermore in this case the set $\mathrm{acc}(a_n)$ has no further elements.
Consequently, for convergent sequences, the set of accumulation points is never empty. In what follows we are going to show that $\mathrm{acc}(a_n)$ is never empty.

• **Theorem (Monotone Subsequence Theorem).** Each real sequence has a monotone subsequence.

• **Corollary (Bolzano–Weierstrass Theorem).*** Each bounded sequence has a convergent subsequence.

• Let (a_n) be a real sequence.

- ⋆ *If* (a_n) *is not bounded:* Then there exists a subsequence which diverges to $-\infty$ or $+\infty$. Indeed, by the fact that the sequence is not bounded it is not bounded from below or from above. If it is not bounded from below then let $b_1 := a_1$ and, for $n \geq 2$, let $b_n := a_k$, where $k := \min\{\ell \in \mathbb{N} \mid a_\ell \leq b_{n-1}\}$. Then (b_n) is a subsequence of (a_n), which diverges to $-\infty$, thus $-\infty \in \mathrm{acc}(a_n)$. The complementary case can be treated similarly.
- ⋆ *If* (a_n) *is bounded:* Then, in view of the Bolzano–Weierstrass Theorem, the set $\mathrm{acc}(a_n)$ has at least one element.

Consequently, for any (a_n), the set $\mathrm{acc}(a_n)$ is indeed nonempty.

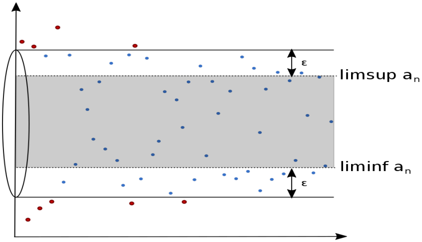
• *A fundamental property of $\mathbb{R}$ is the following:*
If a nonempty subset of $\mathbb{R}$ is bounded from below (or bounded from above) then its infimum (or its supremum) exists.

• **Definition (Lower and upper limit).** Let (a_n) be a real sequence. Then the extended real number

$$\liminf_{n\to\infty} a_n := \inf \operatorname{acc}(a_n), \quad \text{or} \quad \limsup_{n\to\infty} a_n := \sup \operatorname{acc}(a_n)$$

is called the lower limit or the upper limit of (a_n), respectively.

Lower limit and upper limit



⋆ For all (a_n) we have

$$\liminf_{n\to\infty} a_n \leq \limsup_{n\to\infty} a_n.$$

⋆ A real sequence (a_n) is convergent of and only if

$$\limsup_{n\to\infty} a_n \leq \liminf_{n\to\infty} a_n.$$

Then the common value is nothing else but the usual limit of the sequence.

Exercises

• Determine the set of accumulation points of the following real sequences and calculate their lower limit and upper limit! Investigate the following sequences for convergence too!

1. $a_n := (-1)^n$
2. $a_n := 1 + \frac{(-1)^n}{n}$
3. $a_n := \sqrt{n+1}$
4. $a_n := \sin(\frac{n\pi}{2})$

• Do we have a real sequence (a_n) for which $\mathrm{acc}(a_n) = [0,1]$ or $\mathrm{acc}(a_n) =]0,1[$ holds?*

• Do we have a real sequence (a_n) which takes each natural number infinitely many times, more precisely, for which for all $m \in \mathbb{N}$ we have that the set

$$\{n \in \mathbb{N} \mid a_n = m\}$$

has infinitely many elements?

• **Theorem.** Let (a_n) and (b_n) be convergent real sequences tending to the limits $A \in \mathbb{R}$ and $B \in \mathbb{R}$, respectively, and let $\lambda \in \mathbb{R}$. Then the following statements hold.
(1) The sum $(a_n + b_n)$ of (a_n) and (b_n) is convergent and tends to $A + B$ as $n \to \infty$.
(2) The scalar multiple (λa_n) is convergent and tends to λA as $n \to \infty$.
(3) The product $(a_n b_n)$ of (a_n) and (b_n) is convergent and tends to AB as $n \to \infty$.
(4) If, in addition, $b_n B \neq 0$ for all $n \in \mathbb{N}$, then the ratio $(\frac{a_n}{b_n})$ of (a_n) and (b_n) is convergent and tends to $\frac{A}{B}$ as $n \to \infty$.

• Statements (1) and (2) mean that the set of all convergent real sequences forms a *real vector space*.

• **Definition (Null sequence).** If $(a_n) \subseteq \mathbb{R}$ converges to zero, then it is called a *null sequence*.

• **Remark.** If you multiply a real number by zero, the result is zero. The analog of this statement is not necessarily true for real sequences, or more precisely: *if you multiply a null sequence by an arbitrary real sequence, you cannot predict what the product will be in terms of convergence.*

- $\star$ Let $a_n := \frac{1}{n}$ and $b_n := n$ if $n \in \mathbb{N}$. Then $a_n \to 0$ and $b_n \to +\infty$ as $n \to \infty$. However: $a_n b_n = 1 \to 1$ as $n \to \infty$.
- $\star$ Let $a_n := \frac{1}{n^2}$ and $b_n := n$ if $n \in \mathbb{N}$. Then $a_n \to 0$ and $b_n \to +\infty$ as $n \to \infty$. However: $a_n b_n = \frac{1}{n} \to 0$ as $n \to \infty$.
- $\star$ Let $a_n := \frac{1}{n}$ and $b_n := n^2$ if $n \in \mathbb{N}$. Then $a_n \to 0$ and $b_n \to +\infty$ as $n \to \infty$. However: $a_n b_n = n \to +\infty$ as $n \to \infty$.

• **Theorem.** If (a_n) is a null sequence and (b_n) is bounded, then $(a_n b_n)$ is a null sequence. Consequently, the product of a null sequence and a convergent sequence is a null sequence.

• *However, there is something to be said for the unbounded case:* Let $k \in \mathbb{Q}$ and $q \in \mathbb{R}$ with $|q| < 1$ be arbitrarily fixed. Then $(n^k q^n)$ is a null sequence.

Limits of notable convergent real sequences

• Let $k \in \mathbb{Q}$ be arbitrarily fixed. Then

$$n^k \to \begin{cases} +\infty & \text{if } k > 0, \\ 1 & \text{if } k = 0, \\ 0 & \text{if } k < 0. \end{cases}$$

• Let $\lambda \in \mathbb{R}$ with $\lambda > 0$. Then

$$\lambda^n \to \begin{cases} +\infty & \text{if } \lambda > 1, \\ 1 & \text{if } \lambda = 1, \\ 0 & \text{if } \lambda < 1. \end{cases} \qquad \text{and} \qquad \sqrt[n]{\lambda} \to 1$$

as $n \to \infty$. Moreover, we have $\sqrt[n]{n} \to 1$ as $n \to \infty$. On the other hand, the sequence $(\sqrt[n]{n!})$ diverges to $+\infty$.

• The sequences of the form

$$\frac{\text{constant}}{\text{polynomial}}, \quad \frac{\text{polynomial}}{\text{exponential}}, \quad \frac{\text{exponential}}{\text{factorial}}, \quad \text{and} \quad \frac{\text{factorial}}{\text{double exponential}}$$

are null sequences.

• Let $k \in \mathbb{Q}$ be arbitrary but fixed and $(x_n) \subseteq \mathbb{R}$ such that $x_n \to +\infty$ as $n \to \infty$. Then

$$\lim_{n\to\infty} \left(1 + \frac{k}{x_n}\right)^{x_n} \to e^k.$$

• **Theorem.** Let (a_n) and (b_n) be convergent real sequences tending to $A \in \mathbb{R}$ and $B \in \mathbb{R}$, respectively.
(1) If there exists $n \in \mathbb{N}$ such that $a_k < b_k$ for all $k \geq n$, then $A \leq B$ holds.
(2) If $A < B$ holds, then there exists $n \in \mathbb{N}$ such that $a_k < b_k$ for all $k \geq n$.

• **Theorem (Squeeze Theorem).** Let (a_n) and (b_n) be convergent real sequences tending to the same limit $L \in \mathbb{R}$. If (c_n) is a real sequence such that there exists $n \in \mathbb{N}$ with $a_k \leq c_k \leq b_k$ whenever $k \geq n$, then (c_n) is convergent and tends to L as $n \to \infty$.

• *Show that* $c_n := \sqrt[n]{2^n + 3^n}$ *tends to* 3 *as* $n \to \infty$.

$$c_n = \sqrt[n]{2^n + 3^n} \leq \sqrt[n]{3^n + 3^n} = \sqrt[n]{2 \cdot 3^n} = 3 \cdot \sqrt[n]{2} =: b_n,$$
$$c_n = \sqrt[n]{2^n + 3^n} \geq \sqrt[n]{3^n} = 3 =: a_n.$$

The sequences (a_n) and (b_n) are convergent having the same limit $L := 3$. Moreover, we have $a_n \leq c_n \leq b_n$ for all $n \in \mathbb{N}$. In view of the *Squezze Theorem*, (c_n) is convergent and tends to 3 as $n \to \infty$.